B. Tech Degree III Semester Examination November 2011

IT/EC/ME/EB/EI 302 ELECTRICAL TECHNOLOGY

(2006 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) Derive E.M.F. equation of a single phase transformer. Show that emf/turn is same for both primary and secondary windings.

(b) Give a brief description of Power transformers and Distribution transformers and their applications.

(c) A 10kW, 250V, dc, 6 pole shunt generator runs at 1000 rpm, when delivering full load. The armature has 534 lap connected conductors. Full load Cu loss is 0.64 KW. The total brush drop is 1 Volt. Determine the flux per pole. Neglect shunt current.

(d) Determine developed torque and shaft torque of 220V, 4pole DC series motor with 800 conductors wave connected and supplying a load of 8.2KW by taking 45A from the mains. The flux per pole is 25mWb and its armature circuit resistance is 0.6 Ω.

(e) Define
 (i) Pitch factor or Coil span factor
 (ii) Breadth factor or winding factor.

(f) Show the effect of increased load on a synchronous motor under conditions of normal, under and over excitations.

(g) Explain the essential factors which influence the choice of site for a Hydro electric power plant.

(h) Define Corona and Skin effect.

PART B

(4 x 15 = 60)

II. (a) With help of vector diagram, explain the characteristics of a Transformer on load. (7)

(b) Obtain the equivalent circuit of a 200/400V, 50Hz, I-phase transformer from the following test data. (8)
O.C Test: 200V, 0.7A, 70W --- L.V.side
S.C Test: 15V, 10A, 85W --- on H.V side
Calculate the secondary voltage when delivering 5KW at 0.8 power factor lagging, the primary voltage being 200V.

OR

III. (a) Explain the working principle and construction of Auto-transformer. (8)

(b) A 10KVA, 500/250V, 50Hz, single phase transformer has a net area of cross-section 90Cm^2 and maximum flux density is 1.2T. Calculate the number of turns of both primary and secondary. (7)

IV. (a) A 6 pole DC generator has an armature with 90 slots and 8 conductors per slot and runs at 1000 rpm, the flux per pole is 0.05Wb. Determine the induced emf if winding is (i) Lap connected (ii) Wave connected. (8)

(b) Derive the emf equation of DC generator from fundamentals and explain why lap winding is preferred for low voltage high current applications. (7)

OR

V. (a) Define commutation. Explain the process of commutation in DC generators with neat sketches. (8)

(b) The armature of a 6-pole lap wound DC shunt motor takes 400A at a speed of 350rpm. The flux per pole is 80mWb, the number of turns is 600, and 3% of the torque is lost as windage, friction and iron losses. Calculate brake horse power. (7)

(P.T.O)

VI. (a) A 3-phase, star connected, 1000 KVA, 11000V alternator has rated current of 52.5A. The AC resistance of the winding per phase is 0.45 Ω. The test results are given below:
O.C.Test: Field current = 12.5A, Voltage between lines = 422V
S.C.Test: Field current = 12.5A, Line current = 52.5 A
Determine the full load voltage regulation of the alternator at (i) 0.8 pf lagging (b) 0.8pf leading.

(b) What are the causes of harmonics in the voltage wave form of an alternator? How can these be minimized?

OR

VII. (a) Explain effects of varying excitation on armature current and pf in a synchronous motor. Draw 'V' Curves.

(b) The power input to a 500V, 50Hz, 6-pole, 3-phase Squirrel Cage Induction motor running at 975 rpm is 40 KW. The stator losses are 1KW and the friction and windage losses are 2KW. Calculate (i) slip (ii) rotor Cu loss (iii) brake horse power (iv) the efficiency.

VIII. With a neat sketch explain thermal power plant.

OR

IX. (a) Write short notes on:
- (i) Electrical equipments in power stations
- (ii) Circuit breakers

(b) Compare AC and DC transmission systems.

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